

The MOND Acceleration Scale from MeerKAT: 47 Deep-Regime H I Discs in MIGHTEE-HI COSMOS

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Abstract

The programme ties the MOND acceleration scale to the vacuum, $a_0 = \kappa c \sqrt{G \rho_\Lambda}$, with $\kappa = \frac{1}{2}$ fitted, not derived; its footings are canonical $a_0 = 9.3603 \times 10^{-11} \text{ (}\rho_\Lambda\text{)}$ and alternative $1.1312 \times 10^{-10} \text{ m s}^{-2} \text{ (}\rho_{\text{crit}}\text{)}$. We run a framework-native width chain, frozen before any width was read, on the public MeerKAT MIGHTEE-HI COSMOS catalogue (293 H I sources): W_{50} and inclination give V , the baryons are $1.33 M_{\text{HI}} + M_\star$ at half the H I diameter, and a median-residual estimator with the ν_{mono} kernel returns a_0 . 47 golden discs survive, all deep in the MOND regime ($g_{\text{bar}}/a_0 \leq 0.084$) and 37 gas-dominated; four calibration controls pass, among them a SPARC closure of the baryon side (+0.008 dex). The pooled scale is $a_0 = 1.046 \times 10^{-10} \text{ m s}^{-2}$ (bootstrap 68% 1.01–1.15; recipe half-width ± 0.143 dex): +0.048 dex from the canonical footing, −0.034 dex from the alternative and −0.060 dex from SPARC’s 1.20; $\kappa = 0.56 \text{ (}\rho_\Lambda\text{)}$ or 0.46 (ρ_{crit}). The direct $V^4/(GM_b)$ is 1.25×10^{-10} (1.19 for the gas-dominated discs), which includes the kernel’s next-order factor (+0.078 dex). The dominant systematic is the H I flux scale. The released cubes confirm the catalogue widths (−0.022 dex, scatter 0.039 dex) but hold less flux. Arecibo single-dish (ALFALFA) fluxes for 15 overlapping galaxies exceed the catalogue’s by 0.195 dex (the catalogue holds 0.64 of them) and the cubes’ by more, so the cube scale is not supported; if that offset carried over to our fainter, more distant sample, a_0 would fall to about $0.6\text{--}0.75 \times 10^{-10}$. On present evidence the flux scale places a_0 between about 0.6 and 1.05×10^{-10} , the catalogue value at the top; the velocity frame of the catalogue widths could add +0.098 dex. On the catalogue’s flux scale the result is consistent with SPARC and with both footings, on the single-dish scale below both. It does not separate the footings and, at $z \leq 0.093$, cannot test the constant $a_0(z)$ the framework predicts (the rival’s lever is 0.009 dex). The cold mass ($\Omega_c h^2 \approx 0.12$) is still required.

1 Introduction

The relation and its scale. Milgrom’s modified dynamics [M83] replaces the Newtonian acceleration below a scale a_0 . In rotating galaxies the observed centripetal acceleration g_{obs} follows the baryonic one g_{bar} along a single radial acceleration relation: 2693 points in 153 SPARC galaxies [LMS16, MLS16, L17], with SPARC’s fitted scale $g_\dagger = 1.20 \times 10^{-10} \text{ m s}^{-2}$ [MLS16]. A full joint inference over the SPARC distances, inclinations, luminosities and mass-to-light ratios finds $a_0 = (1.19 \pm 0.04 \text{ (stat)} \pm 0.09 \text{ (sys)}) \times 10^{-10} \text{ m s}^{-2}$ [D23].

MeerKAT. The MIGHTEE-HI survey [MM26] measures H I out to $z \approx 0.09$ with one instrument and one pipeline. From its early-science data, Ponomareva et al. [P21] built the H I baryonic Tully–Fisher relation of 67 galaxies from both the global-profile width W_{50} and resolved rotation curves, and found no evolution over $0 \leq z \leq 0.081$. Rajohnson et al. [R22] measured the H I size–mass relation of 204 galaxies (slope 0.501, intercept −3.252). Văraşteanu et al. measured the radial acceleration relation with resolved, SED-based stellar mass-to-light ratios, first in 19 H I-selected galaxies to $z = 0.08$ [V25], then in 130 resolved MIGHTEE-HI and LADUMA galaxies to $z \approx 0.09$ [V26], finding $a_0 = (1.50 \pm 0.05) \times 10^{-10} \text{ m s}^{-2}$ with their kernel and mass modelling and no significant redshift evolution. The MIGHTEE-HI COSMOS catalogue [MM26] lists 293 sources at $0.004 < z < 0.093$ with H I masses, W_{50} widths, inclinations and stellar masses.

The framework. The programme ties the scale to the vacuum energy density, $a_0 = \kappa c \sqrt{G \rho_\Lambda}$, with $\kappa = \frac{1}{2}$ fitted, not derived. With $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $\Omega_\Lambda = 0.6847$ [PL20] this gives the canonical footing $a_0 = 9.3603 \times 10^{-11} \text{ m s}^{-2}$; the same coefficient on the critical density gives the alternative footing 1.1312×10^{-10} . Its galaxy law is a MOND kernel ν with $g_{\text{obs}} = \nu(g_{\text{bar}}/a_0) g_{\text{bar}}$. The framework’s own closed form, $\nu = \sqrt{1 + a_0/g_{\text{bar}}}$, is the relation contained in Milgrom’s vacuum-effect paper [M99] (with a different coefficient); the measurement lanes use the monotone kernel ν_{mono} , which below $g_{\text{bar}}/a_0 = 2.54$ equals the exponential form $\nu = [1 - \exp(-\sqrt{y})]^{-1}$ ($y = g_{\text{bar}}/a_0$) to 2×10^{-4} . That form is the $\alpha = \frac{1}{2}$ member of the family of Milgrom & Sanders [MS08] and the fitting function of [MLS16]. The distinctive prediction is a constant $a_0(z)$; the rival reading $a_0 \propto H(z)$ rises by a factor $E(z)$. The framework adds no dark-matter particle, but the cold mass ($\Omega_c h^2 \approx 0.12$) is still required for cosmology.

This paper. We report a framework-native, catalogue-based measurement of the local a_0 from MIGHTEE-HI (lane CFG301), with criteria frozen and committed before any width, velocity or mass value was read, four calibration controls that could fail, a re-measurement of the inputs from the released cubes (lane CFG302) and an independent check of the H I flux scale against Arecibo single-dish fluxes (lane CFG304). No halo model enters; the only cosmological input is the catalogue’s luminosity distance.

2 Data

The catalogue. The MIGHTEE-HI COSMOS catalogue [MM26] gives, per source: z_{HI} ; the luminosity distance D_L (flat Λ CDM, $H_0 = 70$, $\Omega_M = 0.3$, no peculiar-velocity correction); $\log M_{\text{HI}}$; the 3-D signal-to-noise SN_{3D} ; W_{50} from a busy-function fit [W14] at 50% of the peak, not corrected for inclination; the inclination from g -band isophotes with intrinsic thickness $q_0 = 0.2$; the stellar mass from BAGPIPES SED fits [C18]; and five quality flags (low confidence, blended, confused, bad ellipse, contaminated). The *golden sample* has all five flags zero. The catalogue’s distances agree with flat Λ CDM at $H_0 = 70$ to 0.45% over the survivors, and its inclinations follow from its axis ratios with $q_0 = 0.2$ to 0.8° (rounding; this paper’s script).

The frozen cut. Applied by code, in this order, before any width was read as data: the golden sample (293 to 202 sources); $0.02 \leq z_{\text{HI}} \leq 0.093$ (188; a 300 km s^{-1} peculiar velocity is then at most 5% of the distance); $45^\circ \leq i \leq 80^\circ$ (122); $\log M_{\text{HI}} \geq 9.0$ (78); $W_{50} \geq 80 \text{ km s}^{-1}$ with $\sigma_{W_{50}}/W_{50} \leq 0.15$ (70); $SN_{3D} \geq 8$ (47); finite inputs (47). The 47 survivors are split by redshift into three equal-count windows: W1 (16 galaxies, z 0.0266–0.0476, median 0.042, median D_L 185 Mpc), W2 (16, z 0.0506–0.0756, median 0.067, 300 Mpc) and W3 (15, z 0.0773–0.0930, median 0.081, 368 Mpc). The median gas fraction is 0.70.

3 Method

The chain. For each galaxy,

$$V = \frac{W_{50} - \delta}{2(1+z)\sin i}, \quad M_b = 1.33 M_{\text{HI}} + M_\star, \quad R = \frac{1}{2} D_{\text{HI}}, \quad \log_{10} \frac{D_{\text{HI}}}{\text{kpc}} = 0.506 \log_{10} \frac{M_{\text{HI}}}{M_\odot} - 3.293,$$

$$g_{\text{bar}} = \frac{GM_b}{R^2}, \quad g_{\text{obs}} = \frac{V^2}{R}, \quad D = \frac{g_{\text{obs}}}{g_{\text{bar}}},$$

with $\delta = 0$ (the primary; a turbulence term $\delta = 11 \text{ km s}^{-1}$ is a recipe knob), the catalogue inclination, the factor 1.33 for helium, no molecular gas, and the BAGPIPES $M_\star$. The size relation has the form of Wang et al. [W16]; its coefficients are those adopted in the programme’s earlier width lane (CFG260) and were not re-checked against [W16] here. All baryons are placed inside R as a point mass.

The estimator. The implied ratio s^* is the root of median $_i \log_{10}[D_i/\nu(g_{\text{bar},i}/(a_0 s))] = 0$ with $\nu = \nu_{\text{mono}}$ and a_0 the canonical value; the measured scale is $s^* \times 9.3603 \times 10^{-11} \text{ m s}^{-2}$ (the alternative footing gives the same absolute a_0 to 7×10^{-17} dex). Intervals come from a bootstrap over galaxies ($B = 4,000$). In the deep limit $\nu \rightarrow y^{-1/2}$ and the estimator reduces to $a_0 = V^4/(GM_b)$, independent of R ; at the next order a galaxy on the law shows $V^4/(GM_b) = a_0 y \nu(y)^2 > a_0$, and the estimator removes that factor.

The recipe half-width is the quadrature of four knobs, each applied to every galaxy: $\delta = 11 \text{ km s}^{-1}$; $M_\star \pm 0.25$ dex; D_{HI} (hence R) ± 0.15 dex; and the distances moved from $H_0 = 70$ to 67.4 (the value behind ρ_Λ). The isotropic $\sin 60^\circ$ in place of the catalogue inclinations is reported as a sensitivity. A second statistic, frozen with the first, is the direct ratio $a_0^{\text{BTFR}} = V^4/(GM_b)$ per galaxy, with its median over (i) all survivors, (ii) the gas-dominated discs ($1.33 M_{\text{HI}} > M_\star$) and (iii) the deep discs ($y < 0.5$).

Calibration controls (frozen; evaluated before any a_0 was printed; each could fail).

- *CC1, local level:* the nearest window W1 must return s^* within ± 0.15 dex of the SPARC scale (1.20/0.93603 = 1.282) and of the programme’s ALFALFA control (CFG281, $s^* = 1.449$ [H18, D20]), with a recipe half-width ≤ 0.20 dex. W1 gives $s^* = 1.233$: -0.017 dex from 1.282, -0.070 dex from 1.449, half-width 0.136. PASS.
- *CC2, SPARC closure of the baryon side:* the same M_b (with $0.5 L_{[3.6]}$ for the stars), size relation and point mass applied to 123 SPARC galaxies (quality ≤ 2 , $i \geq 30^\circ$, $V_{\text{flat}} > 0$) with $g_{\text{obs}} = V_{\text{flat}}^2/R$ return $s^* = 1.304$, $+0.008$ dex from 1.282. With the measured R_{HI} in place of the relation the value is 1.324; the relation is unbiased on SPARC (median -0.002 dex). PASS.
- *CC3, drift:* $|\log s^*(\text{W3}) - \log s^*(\text{W1})| \leq 0.10$ dex. The drift is -0.057 dex. PASS, with little power: the statistical width of the drift is 0.136 dex.
- *CC4, self-test:* in 100 worlds with widths fabricated from the law at $s = 2$ on the survivors’ own masses and inclinations (0.15 dex scatter, 10% width errors), the bias is -0.013 dex (tolerance 0.023) and the 95% interval covers the truth in 93 of 100. PASS.

Reactivity and reproduction. Widths multiplied by 1.1892 raise the pooled s^* by $+0.322$ dex (expected 0.30 ± 0.05 ; MUTATE=1); $\sin i_{\text{eff}} = 0.70$ against the $\sin 60^\circ$ variant raises it by $+0.392$ dex (expected 0.37 ± 0.06 ;

set	N	z median	D_L [Mpc]	s^*	a_0 [$10^{-10} \text{ m s}^{-2}$]	68%	95%	recipe [dex]
W1	16	0.042	185	1.233	1.154	1.045–1.312	0.964–1.910	0.136
W2	16	0.067	300	1.089	1.019	0.931–1.153	0.768–1.311	0.139
W3	15	0.081	368	1.081	1.012	0.780–1.336	0.467–1.424	0.135
pooled	47	—	—	1.117	1.046	1.012–1.145	0.936–1.311	0.143

Table 1: The implied a_0 per redshift window and pooled (CFG301; $s^* = a_0/9.3603 \times 10^{-11}$). The 68% and 95% columns are bootstrap intervals over galaxies, in $10^{-10} \text{ m s}^{-2}$; the recipe column is the half-width of the four-knob quadrature.

MUTATE=2). The chain, fed CFG281’s ALFALFA arrays, reproduces CFG281’s committed s^* exactly (1.332408 unmatched, 1.449173 matched; M5). All six frozen hand estimates hit. An independent re-run reproduced every results file identically.

Disclosures. Before the freeze only the catalogue’s header line had been read. The frozen text asks for mass-matched windows in CC1 and CC3 but does not define the matching; the rule fixed in the script before any value was read (W1 and W2 matched to W3’s $\log M_{\text{HI}}$ quartile bins, at least three galaxies per bin) could not be met (W1 had 8/3/2/2 galaxies per bin), so CC1, CC3 and the drift use unmatched windows. W1’s median $\log M_{\text{HI}}$ (9.605) is 0.155 dex below W3’s (9.76). A reader who insists on matching should treat CC1 and CC3 as not evaluable; the calibrated label then rests on CC2 and the unmatched numbers. The BTFR kernel-factor diagnostic was added after the SPARC control and before the measurement, using baryons only.

4 Results

The level. The pooled scale is $a_0 = 1.046 \times 10^{-10} \text{ m s}^{-2}$ ($s^* = 1.117$), with a bootstrap 68% interval of 1.012–1.145 and a 95% interval of 0.936–1.311, and a recipe half-width of 0.143 dex (Table 1, Fig. 1). It lies +0.048 dex above the canonical footing, −0.034 dex below the alternative and −0.060 dex below SPARC’s 1.20. The recipe knobs contribute 0.106 (δ), 0.086 (M_*), 0.026 (D_{HI}) and 0.036 dex (H_0); the $\sin 60^\circ$ sensitivity is 0.026 dex (0.146 with it in the quadrature). All 47 galaxies are deep: $y = g_{\text{bar}}/a_0$ at R has quartiles 0.028, 0.033 and 0.042 and a maximum of 0.084.

The windows and the drift. The window medians are 1.154, 1.019 and 1.012 (Table 1); their bootstrap intervals are lumpy, because a median of 15–16 galaxies jumps between galaxies. The slope of $\log_{10} s^*$ on $\log_{10} D_L$ is −0.20 per dex (68% −0.67 to +0.08; 95% −1.25 to +0.35): not resolved. The ratio W3/W1 is 0.877 ($\log_{10}$: −0.057±0.136). The programme’s ALFALFA control falls by 0.105 dex between 20–50 and 50–85 Mpc (CFG281), the same sign. Between W1 and W3 the rival $a_0 \propto H(z)$ predicts +0.009 dex, far below the statistical width.

The direct BTFR. The median $V^4/(GM_b)$ is $1.251 \times 10^{-10} \text{ m s}^{-2}$ over all 47 (68% 1.187–1.348; +0.126 dex from the canonical footing, +0.044 dex from the alternative; recipe 0.123 dex) and 1.187×10^{-10} over the 37 gas-dominated discs (68% 1.109–1.251; +0.103 and +0.021 dex; recipe 0.128 dex). The deep subset (iii) is the same 47 galaxies as (i). These rows sit above the s^* row by design: computed from the baryons alone, the median kernel factor $y\nu_{\text{mono}}(y)^2$ is 1.197 (+0.078 dex) at the canonical a_0 and 1.172 at 1.20×10^{-10} , and 1.183 for the gas-dominated discs. The gas-dominated median differs from the full one by −0.023 dex, so the stellar mass hardly matters. The same recipe on SPARC gives a BTFR level of 1.526×10^{-10} (CC2’s diagnostic). Fig. 2 shows the 47 discs against the deep-MOND line on both footings.

The coefficient. Read as $\kappa = \frac{1}{2} a_0/a_0^{\text{footing}}$, the pooled scale gives $\kappa = 0.559$ on the ρ_Λ footing (bootstrap 68% 0.540–0.612, 95% 0.500–0.701; scaled by $10^{\pm 0.143}$, 0.40–0.78) and $\kappa = 0.462$ on ρ_{crit} (68% 0.447–0.506, 95% 0.414–0.580; recipe 0.33–0.64). On statistics alone $\kappa = \frac{1}{2}$ sits at the lower edge of the 95% interval on the ρ_Λ footing and inside the 68% interval on ρ_{crit} ; with the recipe width it is inside both. These are on the catalogue’s $H_0 = 70$ distances. On the $H_0 = 67.4$ distances that match ρ_Λ , the scale falls by 0.036 dex and the central values become 0.514 and 0.425 (Section 5).

Against other determinations. The pooled scale is −0.056 dex from the joint inference of [D23] (1.19) and −0.157 dex from the MIGHTEE-HI/LADUMA resolved fit of [V26] (1.50), which uses resolved rotation curves, resolved SED mass-to-light ratios and its own kernel; the authors’ sub-samples span 1.31–1.80 (CFG279). The two MIGHTEE values are not the same measurement: ours places all baryons at $D_{\text{HI}}/2$ and reads the velocity from W_{50} , and the samples overlap only in part (not checked galaxy by galaxy). Which H I flux scale the resolved analysis uses we have not checked.

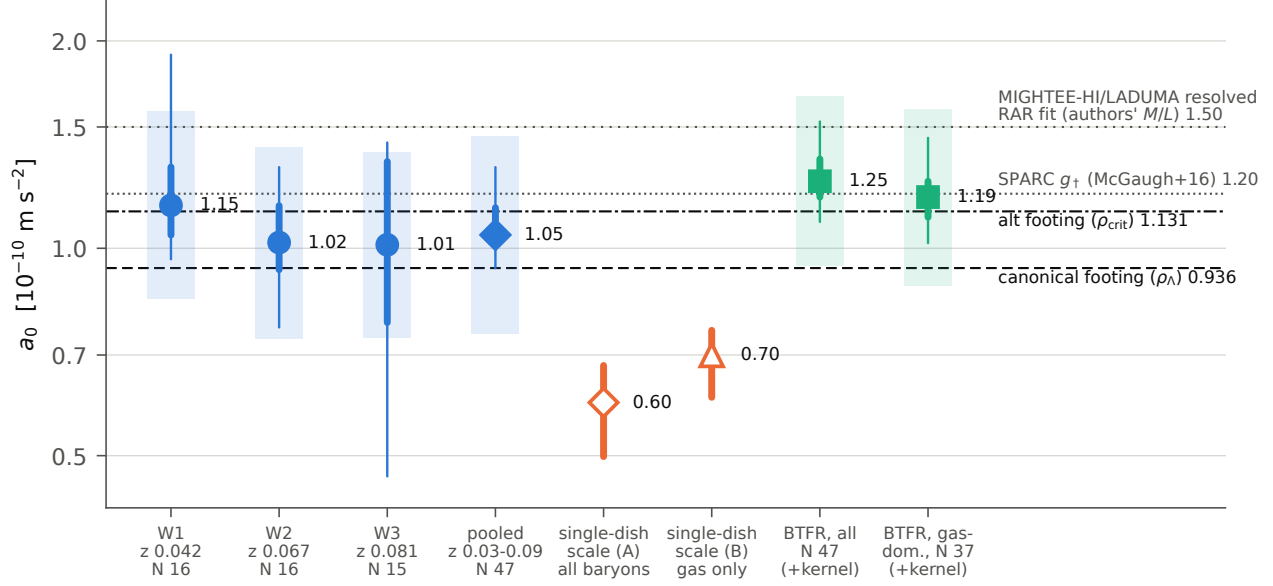


Figure 1: The implied a_0 of CFG301 per redshift window (circles) and pooled (filled diamond), with bootstrap 68% (thick) and 95% (thin) intervals and the recipe half-width (shaded box). Open markers: the pooled value on the Arecibo single-dish flux scale (CFG304’s interpolation of CFG301’s band rows, with the 68% range of its flux ratio), all baryons raised (A, 0.60; 0.50–0.68) or the gas alone (B, 0.70; 0.61–0.76). Squares: the direct BTFR medians, which include the kernel’s next-order factor (+0.078 dex). Lines: the canonical (0.936) and alternative (1.131) footings, SPARC’s fitted $g_{\dagger}$ (1.20) and the MIGHTEE-HI/LADUMA resolved fit with the authors’ mass modelling (1.50), in $10^{-10} \text{ m s}^{-2}$.

5 Systematics

The H I flux scale (CFG302). CFG302 re-measured the 188 golden sources at $0.02 \leq z \leq 0.093$ from the four released MIGHTEE-HI r1p0 cubes (circular restoring beam $74.2\text{--}77.3''$, $8''$ pixels), with criteria frozen before any spectrum was extracted. Of the 179 sources in its primary set, 58 reach its frozen detection threshold. *The widths agree:* the median log ratio of our rest-frame W_{50} to the catalogue’s is -0.022 dex with a robust scatter of 0.039 dex (4 of 58 pull outliers), so the velocity side of the chain stands. *The fluxes do not:* the median ratio over all 179 is 0.22 , and over the 58 detections the median log ratio is -0.30 dex (a ratio of 0.50 , scatter 0.24 dex; 46 of 58 pull outliers). The aperture is complete ($S(2.5\theta)/S(1.5\theta) = 1.002$), and the stacked line holds 0.60 ± 0.05 of the catalogue’s mean flux. One frozen control failed and is kept: the line-free signal-to-noise spread is 0.489 , not about 1, because the noise is anticorrelated over long spectral lags, as if the released spectra were high-pass filtered. For exactly our 47 discs (post hoc join with CFG302’s committed table), 45 are in its primary set and 20 reach its threshold; their median cube-to-catalogue flux log ratio is -0.443 dex, while their widths agree (-0.017 dex).

An independent flux scale (ALFALFA, CFG304). CFG304 matched the catalogue with the ALFALFA catalogue of Arecibo single-dish fluxes [H18], under criteria frozen before any per-galaxy flux was read: 23 matches, 15 of them ALFALFA code 1. The matches are real: catalogue and ALFALFA widths agree to $+0.000$ dex (scatter 0.037), and shuffled positions give 0.275 matches on average. On the 15 code-1 pairs the median log ratio of catalogue to single-dish flux is -0.195 dex (68% -0.250 to -0.156), a ratio of about 0.64 ; for the 7 of them with a cube detection the cube ratio is -0.308 dex. Without pairs that have a catalogued neighbour within $3.5'$ the two ratios are -0.175 and -0.383 . The catalogue’s log M_{HI} sit 0.22 dex below ALFALFA’s for the same galaxies, consistent with the flux ratio. The frozen decision is “neither”: the single dish sees more H I than both MeerKAT scales, and the cube scale lies further from it than the catalogue’s, so the cube scale is not the relevant systematic. The deficit persists at $z \geq 0.02$ (-0.137 dex over 19 pairs, post hoc). Its cause is open: losses in the catalogue’s 3σ masks, the released products, the absolute flux scale, or Arecibo’s $3.5'$ beam adding flux from uncatalogued neighbours.

What the flux scale does to a_0 . The chain uses the catalogue’s M_{HI} . If the single-dish scale applied to our 47 discs, their H I masses would rise by about 0.2 dex and a_0 would fall. CFG304’s interpolation of CFG301’s

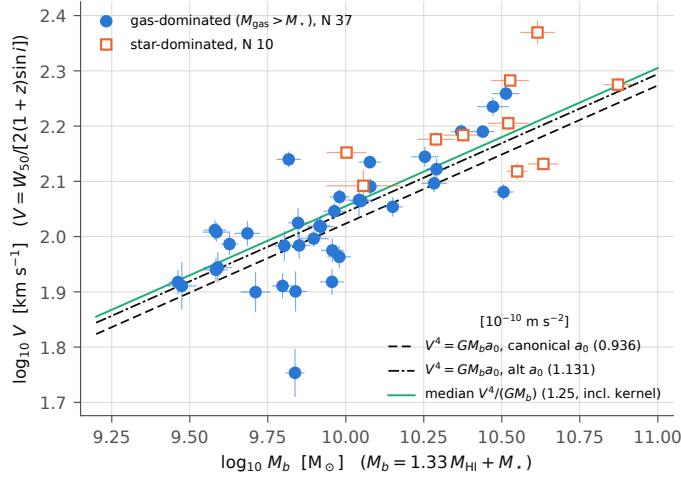


Figure 2: The 47 discs in the plane of $V = W_{50}/[2(1+z)\sin i]$ and $M_b = 1.33 M_{\text{HI}} + M_*$: gas-dominated (filled, 37) and star-dominated (open, 10). Error bars: the catalogue’s W_{50} error and the quadrature of its M_{HI} and M_* errors; inclination errors are not tabulated and not shown. Lines: $V^4 = GM_b a_0$ on the canonical and alternative footings (the deep-MOND limit) and the median $V^4/(GM_b) = 1.25 \times 10^{-10} \text{ m s}^{-2}$, which includes the kernel factor.

variant (all on the same 47 discs)	$\Delta \log_{10} a_0$ [dex]	source
H I flux scale at the single-dish level (catalogue/ALFALFA -0.195 dex): all baryons raised, band interpolation	-0.243	CFG304
same, gas alone (gas fraction 0.70), band interpolation	-0.174	CFG304
same, all baryons raised at fixed R , exact re-run	-0.237	post hoc, this paper
same, gas alone, R from the size relation, exact re-run	-0.144	post hoc, this paper
same, the 7 matched pairs that pass CFG301’s cut (-0.159 dex): all baryons / gas alone	$-0.193 / -0.141$	CFG304 post hoc
velocity frame: catalogue W_{50} read as rest-frame (no $1+z$)	$+0.098$	post hoc, this paper
inclination: $q_0 = 0.10 / 0.30$ in place of 0.2	$+0.025 / -0.055$	post hoc, this paper
inclination: isotropic $\sin 60^\circ$	$+0.026$	CFG301
turbulence $\delta = 11 \text{ km s}^{-1}$	-0.106	CFG301 recipe
stellar mass ± 0.25 dex	$-0.090 / +0.082$	CFG301 recipe
$D_{\text{HI}} \pm 0.15$ dex	$+0.021 / -0.031$	CFG301 recipe
MIGHTEE’s own size relation [R22]	-0.001	post hoc, this paper
distances on $H_0 = 67.4$ instead of 70	-0.036	CFG301 recipe

Table 2: Systematic variants of the pooled a_0 (1.046×10^{-10}). Post hoc rows are this paper’s script (`make_paper40_figures.py`), which first reproduces CFG301’s committed pooled, window, band, H_0 and BTFR values exactly; they were not frozen. The single-dish rows are a signed systematic for galaxies brighter and nearer than ours, not a measurement.

band rows gives 5.97×10^{-11} with all baryons raised (68% 4.98–6.76) and 7.00×10^{-11} with the gas alone at a gas fraction of 0.70 (68% 6.08–7.60). Exact re-runs of the chain (post hoc) give 6.06×10^{-11} (all baryons, fixed R) and 7.50×10^{-11} (the gas alone, each galaxy’s own gas fraction, R following the size relation; 7.11 at fixed R). The 7 matched pairs that pass CFG301’s own cut have a ratio of -0.159 dex, which gives 6.7 and 7.6×10^{-11} (CFG304, post hoc). This is a signed systematic, not a measurement: the matched galaxies are the 23 H I-brightest COSMOS sources at $z \leq 0.047$, while our 47 reach $z = 0.093$, mostly beyond ALFALFA, so whether the offset carries over is an assumption. On present evidence the flux scale places a_0 between about 0.6 and 1.05×10^{-10} , with the catalogue-scale value at the top. On the single-dish scale κ would be 0.32–0.40 (ρ_Λ) or 0.26–0.33 (ρ_{crit}), below $\frac{1}{2}$ on both footings, and a_0 would sit 0.20–0.30 dex below SPARC’s 1.20.

The velocity frame. The chain divides W_{50} by $1+z$, reading the catalogue widths as observed-frame. CFG302’s raw widths agree with that reading to $+0.002$ dex and with the rest-frame reading to -0.022 dex, which does not decide the convention (± 0.025 dex). If the catalogue widths were already rest-frame, the pooled a_0 would rise by 0.098 dex to 1.31×10^{-10} (post hoc; median $z = 0.065$).

Inclinations, turbulence, stellar mass and size. The catalogue’s inclinations follow from its axis ratios with $q_0 = 0.2$; $q_0 = 0.10$ or 0.30 moves the pooled a_0 by $+0.025$ or -0.055 dex (post hoc) and the isotropic $\sin 60^\circ$ by $+0.026$ dex, and the 45° cut keeps the face-on end, where $\sin i$ errors are largest, out. No turbulence correction is applied; $\delta = 11 \text{ km s}^{-1}$, the largest recipe term, lowers a_0 by 0.106 dex. The stellar mass enters weakly because 37 of the 47 discs are gas-dominated (± 0.25 dex on M_* : $-0.090 / +0.082$ dex). In the deep limit the estimator does not depend on R , so the size relation enters only at next order: $D_{\text{HI}} \pm 0.15$ dex moves a_0 by 0.026 dex (half-width), and MIGHTEE’s own relation [R22] by -0.001 dex (D_{HI} smaller by 0.008 dex at the median $\log M_{\text{HI}} = 9.73$). The

point mass at $D_{\text{HI}}/2$ is a recipe choice that CC2 tests on SPARC, not a resolved mass model.

Distances and H_0 . The catalogue’s distances assume $H_0 = 70$ and no peculiar velocities, while ρ_Λ uses $H_0 = 67.4$. Moving the distances to 67.4 raises every mass by $2\log_{10}(70/67.4)$ and lowers the pooled a_0 by 0.036 dex, to 9.62×10^{-11} ($s^* = 1.028$). This is a convention, not an error; quoted alone, the result is on the catalogue’s distances. Peculiar velocities are random per galaxy and at most 5% in distance above $z = 0.02$.

Mass matching and sample size. The windows could not be mass-matched (Section 3), so the drift compares W1, whose median $\log M_{\text{HI}}$ is 0.155 dex lower, with W3. With 15–16 galaxies per window the window medians are lumpy (W3’s 95% interval spans a factor of three), CC3 has little power and the drift slope is unresolved. The sample is HI-selected and mass-limited at high z .

6 What the result can and cannot say

- **It can say:** an independent instrument and sample, with no halo model and a chain calibrated on SPARC, give a deep-regime scale of $1.046 \times 10^{-10} \text{ m s}^{-2}$ on the catalogue’s flux scale, the top of the flux-scale range. That is consistent with SPARC (-0.060 dex) and with both of the framework’s footings, and $\kappa = \frac{1}{2}$ lies inside the recipe width on both.
- **It is not a footing discriminator.** The footings differ by 0.082 dex; the result sits between them, $+0.048$ and -0.034 dex away, with a recipe half-width of 0.143 dex and a single-dish flux systematic of -0.14 to -0.24 dex.
- **It is not an $a_0(z)$ test.** At $z \leq 0.093$ the rival $a_0 \propto H(z)$ differs from a constant by 0.009 dex between W1 and W3, against a statistical width of 0.136 dex. The distinctive constant- $a_0(z)$ prediction needs $z \gtrsim 2$ and baryon calibrations near 0.1 dex [P38].
- **It is conditional on the flux scale.** If the single-dish offset applies to our discs, the same chain gives 0.60 – 0.75×10^{-10} , below both footings and 0.20 – 0.30 dex below SPARC, an offset that the chain’s SPARC closure (CC2, $+0.008$ dex) does not show. Whether it applies to galaxies fainter and more distant than the matched ones is open; it needs single-dish or total-power fluxes for the CFG301 sample itself.
- **Not claimed:** nothing here says the data favour the framework over Λ CDM or over standard MOND; $\kappa = \frac{1}{2}$ is fitted, not derived; the theory is not claimed closed; the cold mass is still required and no dark-matter particle is introduced.

7 Conclusions

On 47 golden, deep-regime ($y \leq 0.084$) MIGHTEE-HI discs, 37 of them gas-dominated, a frozen, calibrated, framework-native width chain gives $a_0 = 1.046 \times 10^{-10} \text{ m s}^{-2}$ (68% 1.01–1.15; recipe ± 0.143 dex) on the catalogue’s flux scale, between the canonical and alternative footings; $\kappa = 0.56$ (ρ_Λ) or 0.46 (ρ_{crit}). The raw cubes confirm the catalogue widths. Arecibo single-dish fluxes for 15 bright, nearby galaxies exceed the catalogue’s by 0.195 dex; if that offset carries over, a_0 falls to 0.60 – 0.75×10^{-10} . Total-power fluxes for the CFG301 discs themselves would settle the flux scale, and the velocity frame of the catalogue widths ($+0.098$ dex if rest-frame) should be confirmed with the catalogue’s authors. The result neither separates the framework’s footings nor tests $a_0(z)$.

Data availability

All scripts, outputs and frozen criteria are in the programme’s public repository. The lanes are `campaign_fresh_gravity/CFG301_mightee_hi_catalogue_width_chain/` (criteria commit 2555ab142, results 6b10c01c0), `campaign_fresh_gravity/CFG302_mightee_cube_raw_widths/` (criteria 7d317dd4f, results a89ba33b9) and `campaign_fresh_gravity/CFG304_mightee_flux_scale_alfalfa/` (criteria 336b36f8f, results 34e40dac6). CFG304 reads the committed ALFALFA tables [H18, D20] in `data_assembly/alfalfa_sdss_local_control/`. The catalogue is the MNRAS supplementary file of [MM26], downloaded by the author in a browser because the publisher’s bot challenge blocks scripted fetches (zip sha256 prefix eb247d5de07eb30e) and committed unmodified as `data_assembly/mightee_hi_catalogue_2026-10-02/MIGHTEE_HI_COSMOS_catalogue.csv` (75,999 bytes, sha256 bcf9e8558bc5644805b9acb288372c17daffac76d8be6c89d7db9843af9fbf23; commit 19adffd49). The MIGHTEE-HI DR1 cubes are at the SARAO DOI 10.48479/jkc0-g916 and are not in the repository. This paper’s script, post hoc numbers and audit are in `qwen_claude_field_theory/papers_2026/`.

Use of AI tools

Large language models from several providers assisted with derivations, data handling, code and drafting: Anthropic’s Claude (mainly via Claude Code), OpenAI models, DeepSeek, Zhipu AI’s GLM, Alibaba’s Qwen and Google’s Gemini. The author reviewed all of the content and takes responsibility for it.

Reproducibility

`make_paper40_figures.py` re-implements the CFG301 chain and writes nothing unless it reproduces CFG301’s committed pooled and window s^* , band row, H_0 knob and BTFR medians to 10^{-9} (it reproduces them exactly). Every number carried by an audit row of `PAPER40_audit.py` is re-read from a committed file at git HEAD or, for the post hoc rows, from `PAPER40_figures_numbers.json`, and is checked to be printed in the tex block carrying its tag; two mutation modes must fail. Ten literature values that no committed file carries are checked only for being printed; numbers quoted only in running text and the references’ journal data are not checked by the script.

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